

Ruisheng Zhang

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RESEARCH INTERESTS

My current research focuses on human attention modeling and display design in extended reality environments. My work lies at the intersection of human factors, design, and engineering psychology.

Keywords: *Human Factors, Attention, Human-Computer Interaction, Human-Automation Interaction*

EDUCATION

University of Michigan, Ann Arbor, MI, USA	08/2025 – Present
<i>Ph.D. Student in Industrial and Operations Engineering (Human Factors)</i>	<i>GPA: 4.0 / 4.0</i>
- Research Direction: Human Attention Modeling, Human-Automation Interaction, Augmented Reality - Advisor: Amelia C. Warden	
Southeast University, Nanjing, CN	09/2022 – 06/2025
<i>M.S. in Design</i>	<i>GPA: 3.86 / 4.0</i>
- Thesis: Research on Hierarchical Human-Machine Interaction Intention Recognition and Prediction Methods - Advisor: Xiaozhou Zhou	
Southeast University, Nanjing, CN	09/2018 – 06/2022
<i>B.E. in Mechanical Engineering</i>	<i>GPA: 3.72 / 4.0</i>
Thesis: Design and Development of Desktop Gesture Interaction System based on Virtual Reality	

RESEARCH EXPERIENCE

The Warden Lab, University of Michigan	08/2025 – Present
<i>Graduate Student Research Assistant (PI: Amelia C. Warden)</i>	
- Conduct research on human attention modeling and attentional guidance in augmented reality (AR) and virtual reality (VR) environments. Use experimental design, task performance metrics, and eye-tracking-based data analysis to understand human attentional mechanisms and inform display design. Human Attention Modeling (Ongoing Project): Investigate how attention distribution changes when using emerging display technologies (such as AR). Extend existing attention models by incorporating additional influencing factors, and develop context-specific predictions of human attention allocation.	
Advanced Human-Computer Interaction Lab, Southeast University	07/2021 – 06/2025
<i>Research Assistant (PI: Xiaozhou Zhou)</i>	
- Focused on system, design, and human factors issues in advanced human-computer interaction (HCI), with research in two main areas: HCI Research in Emerging Display Technologies (AR/VR): (1) Led the development of a VR desktop gesture interaction system (BeyondDeskVR) to reduce arm fatigue associated with mid-air gestures. (2) Contributed to multimodal interaction design for intelligent vehicle cockpits and mid-air gesture interaction in VR sandbox. HCI Research in Complex Dynamic Systems: Led the development of methods for human intention recognition and prediction in human-automation teams: (1) For a complex cockpit task, conducted task analysis and collected user behavioral datasets. (2) Applied neural networks and probabilistic graphical models to achieve multi-level intention recognition and prediction. (3) Enabled intelligent adaptive interface design.	

AWARD & HONORS

• IOE Departmental Fellowship University Of Michigan	2025-2026
• Outstanding Graduate Students (Top 5%) Southeast University	2025
• Southeast University Outstanding Master's Thesis Southeast University	2025
• Suzhou Industrial Park Scholarship (\$500) Southeast University	2023
• The First Prize Graduate Scholarship (\$1,500 / Year) Southeast University	2022-2025
• Outstanding Undergraduate Students (Top 10%) Southeast University	2022
• Southeast University Outstanding Undergraduate Thesis Southeast University	2022
• Min Yu Alumni Award Fund (\$500) Southeast University	2021

PUBLICATIONS

J=Journal, C=Conference, T=Thesis, P=Patents

PEER-REVIEWED JOURNAL PAPERS

- [J1] Wang, B., **Zhang, R.**, Zhu, J., & Zhou, X*. (2026). Visual Search Interference from the Eyes' Effector Role in VR Selection: The Impact of Pointing Modality and Information Density. *International Journal of Human-Computer Interaction*, 1-14. <https://doi.org/10.1080/10447318.2026.2621283>.
- [J2] **Zhang, R.**, Qiu, X., Han, J., Wu, H., Li, M., & Zhou, X*. (2025). Hierarchical Intention Recognition Framework in Intelligent Human-Computer Interactions for Helicopter and Drone Collaborative Wildfire Rescue Missions. *Engineering Applications of Artificial Intelligence*, 143, 110037. <https://doi.org/10.1016/j.engappai.2025.110037>.
- [J3] **Zhang, R.**, & Zhou, X*. (2025). BeyondDeskVR: an Extended Virtual Hand Interaction System in Virtual Reality. *Behaviour & Information Technology*, 44(17), 4264-4285. <https://doi.org/10.1080/0144929X.2025.2469669>.
- [J4] **Zhang, R.**, Wang, B., Zhu, J., & Zhou, X*. How Early Can We Know? Temporal Mechanisms of Predicting Explicit and Implicit Intentions in Adaptive Human-Machine Interaction Systems. **To Be Submitted**.

PEER-REVIEWED CONFERENCE PROCEEDINGS

- [C1] Warden, A.C.*, & **Zhang, R.** (2026). Visual Guidance in Dynamic Search: How Augmented Reality Cue Properties Shape Performance. In *Proceedings of the Human Factors and Ergonomics Society Annual 70th Meeting*. **Under Revision**.

OTHER PUBLICATIONS

- [T1] **Zhang, R.** (2025). Research on Hierarchical Human-Machine Interaction Intention Recognition and Prediction Methods. Master's Thesis. Southeast University.
- [T2] **Zhang, R.** (2022). Design and Development of Desktop Gesture Interaction System based on Virtual Reality. Bachelor Thesis. Southeast University. [PDF].
- [P1] **Zhang, R.**, Zhou, X., Zong, C., Xue, C., & Niu, Y. Desktop Gesture Interaction System based on Virtual Reality. *Chinese Invention Patent*. ZL202210485007.1. [PDF].
- [P2] Zhou, X., **Zhang, R.**, Qiu, X., & Han, J. A Multi-Level Human-Computer Interaction Intention Recognition Method for Complex Task Scenarios. *Chinese Invention Patent*. ZL202411181378.6. [PDF].

IN-PROGRESS MANUSCRIPTS

W=Working Papers

- [W1] Warden.A.C.*, **Zhang, R.** AR-SEEV : Extending the SEEV Framework for Modeling Human Attention in Augmented Reality using a Weather Forecasting Task. **In Progress**.
- [W2] **Zhang, R.**, Warden.A.C.* A Literature Review on Augmented Reality for Space Exploration: Redefining the Future of the Space Workforce. **In Progress**.
- [W3] **Zhang, R.**, Warden.A.C.* A Literature Review on Designing Mixed Reality Systems for Blind and Visually Impaired People with External Devices and Multimodal Cues. **In Progress**.

CONFERENCE PRESENTATIONS & INVITED TALKS

L=Lecture, P=Poster, T=Talk

- [P1] **Zhang, R.**, Qiu, X., Han, J., Wu, H., Li, M., & Zhou, X*. (December, 2024). Hierarchical Intention Recognition Framework in Intelligent Human-Computer Interactions for Helicopter and Drone Collaborative Wildfire Rescue Missions. Poster presentation at the *2nd Academic Conference on Human-Centric Manufacturing in China*.

PROFESSIONAL SERVICE

- **Journal / Conference Reviewer** 2025-Present
Engineering Applications of Artificial Intelligence (EAAI), Behaviour & Information Technology (BIT)

ACADEMIC ACTIVITIES AND SERVICE

- **WISE-GISE Summer Camp** | University of Michigan 2026
Serve as a student leader, contributed to planning and organizing an introductory human factors session and guided 19 middle school students in a design activity on “Designing Mixed Reality with Humans in Mind”.
- **Engage Detroit Workshop** | University of Michigan 2026
Serve as a student volunteer, guiding 42 middle school students participating in a VR experience workshop on “Cognitive Engineering with Virtual Reality”.

TEACHING & MENTORING EXPERIENCE

- **IOE 436: Human Factors in Computer Systems (GSI)** | University of Michigan Winter 2026
- **Undergraduate Students Mentoring** | Southeast University 2020-2021

TECHNICAL SKILLS

Programming	C#, Python, R, C++ , Lua
Languages	Chinese (Native), English (Fluent)
Software	Unity, Figma, Blender, SolidWorks, SPSS, E-Prime, OpenCV
Tools	Machine Learning & Data Processing (PyTorch, Scikit-learn, Pandas, Matplotlib, NumPy) XR Development (Meta XR All-in-One SDK, Oculus Integration, SteamVR, MRTK, Vuforia) Version Control & Collaboration (Git, Github Desktop, SSH)